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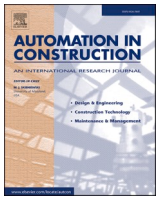

Enhanced real-time detection transformer (RT-DETR) for robotic inspection of underwater bridge pier cracks

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ABSTRACT

The inadequate visual environment reduces the accuracy of underwater bridge pier fracture detection. Consequently, this paper suggests enhancing the backbone of the Real-Time Detection Transformer(RT-DETR) model to serve as the backbone of the YOLOv8 model. This will be achieved by substituting the Faster Implementation of CSP Bottleneck with 2 convolutions(C2f) module with the Poly Kernel Inception(PKI) Block, which is composed of the PKI Module and Context Anchor Attention(CAA) Block. Its strong capability to distinguish cracks and background features enables accurate recognition of underwater bridge pier cracks. To provide data for detecting these cracks, the enhanced Unpaired Image to Image Translation(CycleGAN) network converts land-style bridge crack images to underwater-style fracture images. The proposed model achieved an F1 score of 0.85 and a mAP50 of 0.84. The real-time detection of underwater bridge fractures by the underwater robot was facilitated by the FPS index of 87.47, which optimizes the detection efficiency.

1. Introduction

The safety of the vast number of bridges worldwide is a top priority [1–3]. Additionally, the structure is highly susceptible to damage due to the fact that certain lower portions of the bridge piers are situated in underwater environments. The bridge pier will develop unavoidable cracks and other defects as a result of the vibration induced by traffic flow [4,5], wind and rain erosion [6–8], and accidental ship impact [9]. Professional divers donning frogman work suits are the primary individuals who observe the traditional maintenance method of detecting cracks through the naked eye [10]. This method is inefficient, time-consuming, and poses significant safety risks. Additionally, it is contingent upon the subjective experience of the frogman.

The temperature tracing method is a conventional approach that is employed to facilitate the detection of cracks in underwater concrete buildings. It is based on the theory of heat conduction and identifies cracks by comparing the cooling rate of the heat source between the breach and the surrounding medium. Chen et al. [11] developed a temperature tracer system that included a sensing heating system, a monitoring tube, and a porous casing. They employed numerical simulations to examine the impact of the casing size, porosity, and thermal conductivity on the monitoring results, resulting in an enhanced crack identification procedure. An innovative fracture detection method for underwater concrete structures was developed by Chen et al. [12]. This method monitors the heat dissipation rate in the cracked region by analyzing changes in leakage and convective heat transfer effects caused by irrigation. The crack discrimination index psi was introduced, and the relationship between fracture characteristics and psi was experimentally investigated. Integrating the heating, temperature measurement, and irrigation systems provides an effective technical means for underwater crack identification. In order to increase the difference in the cooling rate of the heat source before and after the fracture appeared, Zhang et al. [13] employed hollow casing. They then employed an enhanced monitoring system to measure the cracks' width. Nevertheless, these methods necessitate the pre-burying of a substantial number of inspection tunnels in the submerged concrete structure, which may impact the performance of the submerged bridge pier construction.

Visual perception is limited in the underwater environment due to its complexity. To address this issue, some scholars employ hydroacoustics to detect cracks in underwater bridge piers. With the DB-VI three-dimensional velocity vector sonar measurement instrument, Wu et al. [14] identified fracture leakage channels in the concrete spillway dam of Banqiao Reservoir and quantified the magnitude and distribution of acoustic energy transfer in the fluid. Hou et al. [15] developed a rapid

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underwater detection framework that employed a deep convolutional neural network and a sonar device to provide quantitative measurements of scour depth and apparent damage. They also employed an enhanced U-Net architecture for semantic segmentation of underwater cracks. However, sonar is a costly method for detecting underwater bridge pier fractures and is easily affected by other underwater factors. It is typically used as an auxiliary device.

The accuracy of crack detection in bridge piers has been significantly improved by the swift advancement of computer technology in recent years. An underwater concrete fracture detection algorithm based on fractal theory was proposed by Yu et al. [16]. This algorithm employs computer vision and digital image processing technology to identify underwater concrete cracks and integrates fractal theory to aid in crack classification. Qi et al. [17] proposed a three-step method for the automatic detection of small fractures during operation, utilizing image enhancement, image slicing input to the convolutional neural network for fracture identification and boundary localization, and subsequently the Otsu algorithm to accurately segment and locate the crack image. In order to facilitate the monitoring of the fracture width and depth of underwater concrete structures, Zhang et al. [18] developed an active sensing technology that is based on embedded smart aggregate. The damage index was established based on the crack depth of underwater concrete beams. A migration learning method known as MA-AttUNet was proposed by Cao et al. [19]. The method employs a multi-level adversarial transfer network to facilitate the transmission of knowledge regarding the features of crack images. The prior knowledge acquired from the source domain can be utilized to segment underwater fissure images. Using the alternating current field measurement (ACFM) technique, current perturbation theory, and gradient imaging algorithm, Yuan et al. [20] developed an intelligent identification method for underwater structural defects. The processed images were accurately identified by the convolutional neural network (CNN) deep learning algorithm. Shi et al. [21] introduced MCL-CrackNet, an innovative crack detection network that improves spatial perception by incorporating a coordinate convolution module and a position attention gate (PAG) module to enhance the recognition of fracture location information. The detection accuracy is enhanced by the network's implementation of a multilevel comparative learning strategy, which effectively extracts and fuses local and global features. The development of computer-based technology is impeded by the scarcity of underwater concrete crack data, and the collection of batch data necessitates a significant quantity of economic, time, and personnel costs.

Underwater robots (ROVs) play a significant role in underwater resource development and exploration [22]. Compared to conventional methods, their efficient data collection significantly reduces the time required to gather underwater bridge pier crack data. These robots are increasingly used across various industries. Simultaneously, the underwater robot can implement deep learning algorithms to enhance the precision of underwater bridge pier fracture detection. Additionally, its long-distance video mapping function [23] and safety and convenience are significantly superior to conventional detection methods. Through the integration of an underwater robot and binocular camera, Ma et al. [24] implemented a ROV-based binocular vision system for the purpose of real-time fissure detection and measurement. In spite of this, the algorithm's detection accuracy is subpar. Cao et al. [25] enhanced the As-Projective-As-Possible (APAP) algorithm for extracting and splicing keyframes from ROV video footage. The spliced fissure image is segmented using a Graph Convolutional Neural Network (GCN). Nevertheless, the segmentation results are not entirely adequate. A fuzzy predictive control algorithm that integrates fuzzy control and model predictive control was introduced by Tian et al. [26]. The algorithm is capable of dynamically adjusting the power output of the propeller in real time in accordance with the dark current data measured by the sensors and performing differential correction until the ROV reaches a predetermined position. This inspection model outperforms the human eye in observing underwater bridge defects mapped by the robot. Wu et al. [27] conducted a crack data collection using a ROV and proposed a real-time, pixel-level crack detection method that is lightweight and real-time. The method is based on an improved instance segmentation model and includes a lightweight backbone and a new efficient prototype mask branch. These features are designed to reduce the complexity of the model and maintain its accuracy. At 129 frames per second (FPS), the proposed method attains an accuracy of 0.945. Nevertheless, the model's robustness requires enhancement, as the dataset is relatively limited. Yu et al. [28] introduced a method for underwater fracture detection and 3D structural appearance reconstruction that employs an underwater wall climbing robot. The structural health was comprehensively demonstrated through semantic segmentation, geometric quantization, and 3D reconstruction, while the integrity of the cracks was guaranteed through a label propagation algorithm. However, the underwater wall-climbing robot's detection efficiency was low. The actual structural surface was restored.

Zhang et al. [29] proposed a real-time automatic segmentation framework for underwater dam cracks, combining LinkNet with a two-stage hybrid migration learning approach. This framework effectively reduces the modeling cost, improves the recognition capability, achieves high-precision segmentation in complex underwater environments, and quantitatively analyzes the cracks in conjunction with an infrared laser system. The framework is based on ROV. Nevertheless, the quality of the underwater images and recordings that have been acquired is subpar. Ye et al. [30] proposed an ultra-lightweight, two-stage damage detection model for underwater structures based on deep learning algorithms. This model addresses the challenges of underwater structure detection by using CycleGAN and MSR to generate defect images, followed by defect detection in the second stage with the lightweight YOLOv5 model. The model significantly improves the detection rate of underwater damage in turbid water environments. However, the methods of CycleGAN and MSR are more complex, which can be addressed by enhancing CycleGAN alone to improve the data effect.

In order to resolve the aforementioned issues, this paper suggests the implementation of an underwater bridge pier crack detection algorithm (UCRT-DETR) in conjunction with an underwater robot to identify underwater bridge pier cracks in real time. Initially, the complexity of the underwater environment and the constraints of the RT-DETR detection model are examined. To accommodate the situation in which the cracks and walls of the underwater piers exhibit less variability, the RT-DETR algorithm's backbone is replaced with the YOLOv8s backbone. The PKI module, which is adept at inspecting slender objects, replaces the C2f module in the YOLOv8 backbone. Secondly, the underwater robot will be employed to detect pier fractures, resulting in the acquisition of a substantial quantity of pier wall image data. The improved CycleGAN model is employed to migrate the manner of underwater pier wall images and land concrete pier crack images, obtain the underwater pier crack data, and provide training support for the detection model, in light of the scarcity of underwater pier crack datasets. In order to evaluate the specific effects of the enhanced detection method and style migration method on the overall performance of the model, we compare it to the YOLOv5s, YOLOv6s, YOLOv8s, and RT-DETR detection models and ablation experiments that are frequently employed. Ultimately, we validate the real-time pier crack detection performance with an underwater robot in a real engineering scenario, thereby showcasing the benefits of the proposed method.

2. Bridge pier crack detection model

Underwater bridge pier crack detection is different from land crack detection, where the land crack is in the air for detection, and the camera can capture good information about the land crack and the area around the crack. The crack itself is often dark in color, and there is some distinction between the wall color around the land crack and the color of the crack itself, which is significantly different. However, in the underwater environment, the longer wavelength of the sun's red light is

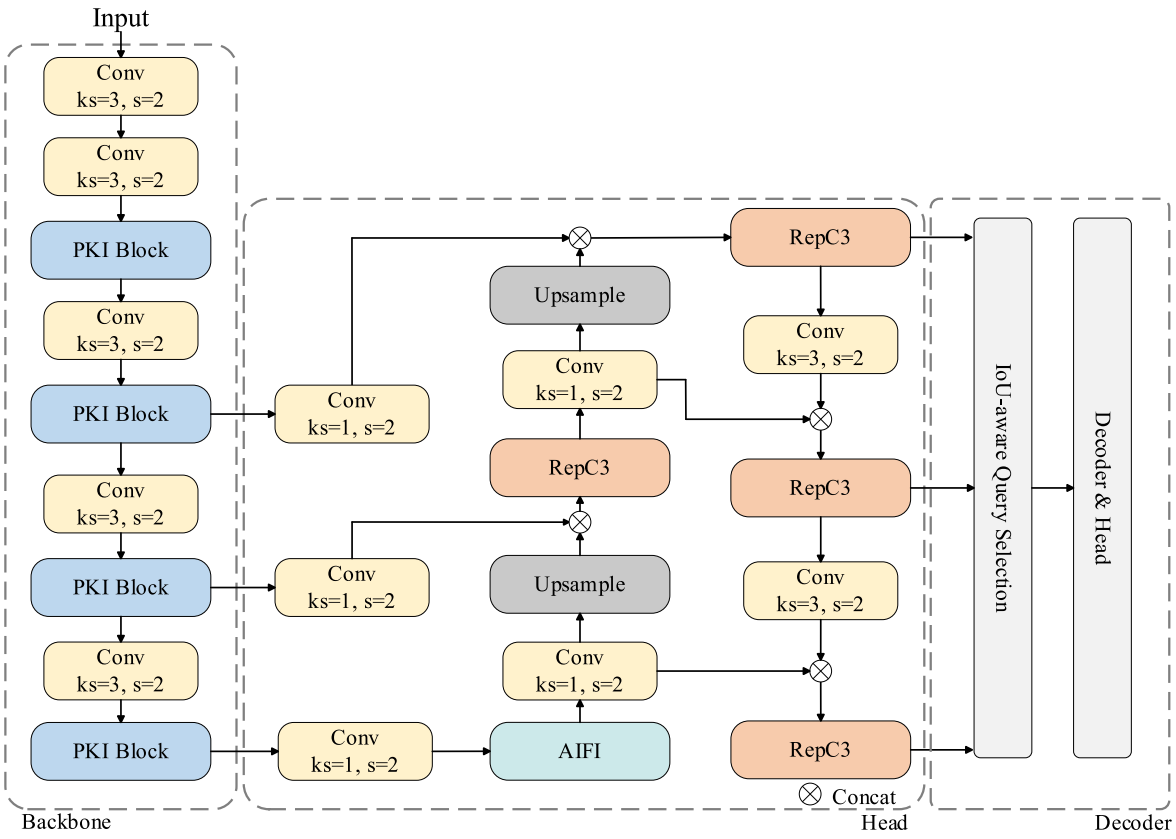

Fig. 1. UCRT-DETR network diagram.

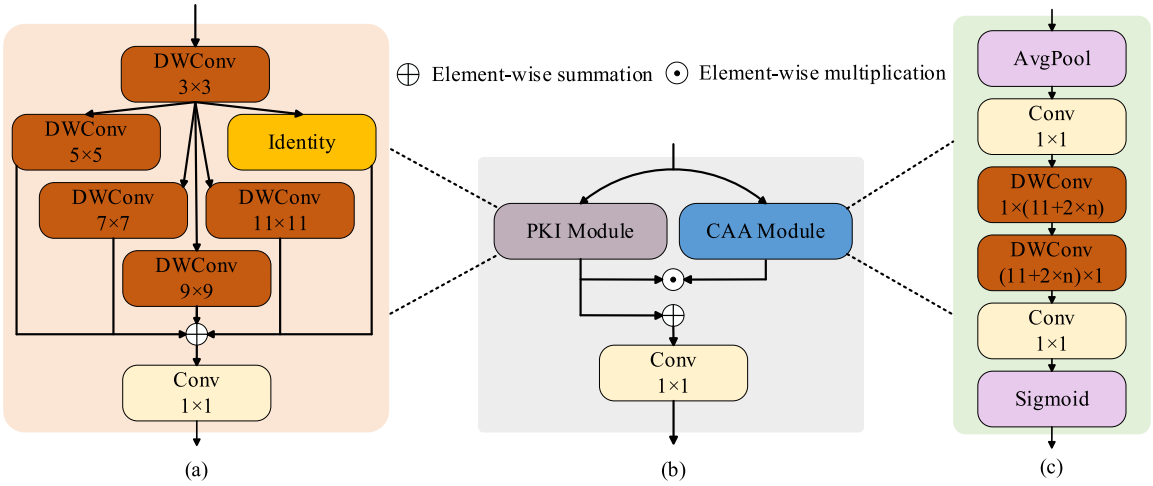

Fig. 2. (a)PKI Module , (b)PKI Block , (c)CAA Module

partially absorbed, resulting in the water body showing blue, green and other dark colors, underwater piers are covered by the water body, the surface of the blue-green and other dark colors, which results in a low degree of differentiation between the cracks in the underwater pier and the wall around the cracks, which is difficult to detect. This is the most difficult challenge in the detection process, and secondly, the impurities in the water will cause the imaging effect of the camera to show a foggy situation, which is also a reason for the low accuracy of underwater bridge pier crack detection. So the underwater bridge pier crack detection method under complex water bodies is proposed. In this section, the overall framework of the proposed modeling approach as well as the improvement strategy for the difficult challenge of bridge pier crack detection under complex water bodies are presented.

2.1. Overall structure

When detecting cracks in underwater bridge piers, there are many uncertainties that cause detection difficulties, firstly, the light conditions underwater are different from those on land, which affects visibility. Secondly, there are underwater currents, underwater robots in the detection process is easy to be disturbed, may lose information. Finally, cracks exist in various forms and sizes.

RT-DETR [31] is an end-to-end target detection model. It is intended to circumvent the non-maximum suppression (NMS) phase in the

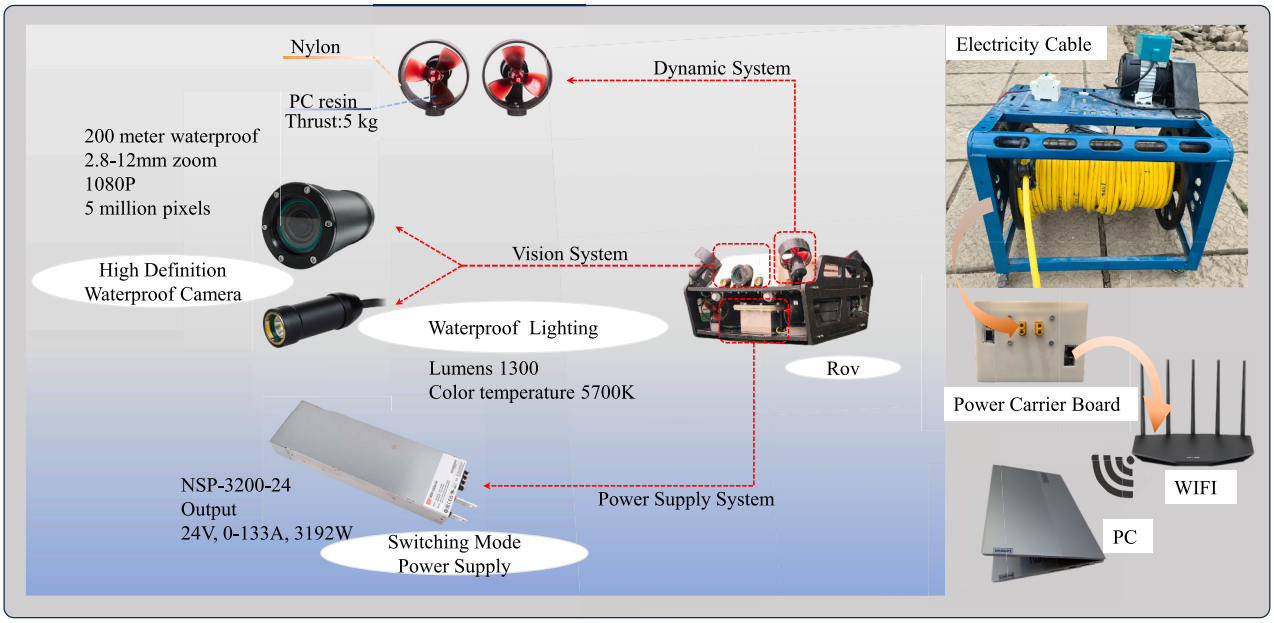

Fig. 3. Overall structure of an underwater robot.

conventional target detection process. The mechanism within the model is optimized by RT-DETR, thereby enhancing the efficiency and speed of detection. It enables fast detection and is suitable for underwater robots to practically detect underwater bridge pier cracks. RT-DETR handles multi-scale features by decoupling intra-scale interactions and cross-scale fusion, which helps to detect underwater bridge pier cracks of different sizes. The efficient hybrid encoder of RT-DETR can adapt to the complex water environment changes. The IoU-aware query selection of RT-DETR can also adapt to such changes and improve the detection performance under different underwater visual perception conditions. Therefore, RT-DETR is selected as the base model to detect underwater bridge pier cracks. RT-DETR using ResNet [32] as its backbone. Efficient Hybrid Encoder and IoU Sense Query Selection Adapts to Complex Water Environment Changes. However, the differences in the feature extraction of its backbone for cracks and for the background of the underwater pier walls are small. The YOLOv8 model's capacity to extract image features is improved, thereby increasing the accuracy of detection, as a result of its backbone design. At the same time inspired by Cai et al. [33] the PKI Block is introduced to replace the C2f module in the YOLOv8 backbone as the new backbone of RT-DETR. This is done to obtain multi-scale texture feature information of the cracks in the underwater bridge pier. The network structure is illustrated in Fig. 1, and it is used to detect the cracks in the wall surface of the underwater bridge pier.

2.2. PKI module

The underwater environment is highly complex, with turbid water and debris around the pier walls. Additionally, the sediment generated by the robot propeller weakens the HD camera's ability to capture detailed information about the pier cracks and walls. The PKI Module [34,35], shown in Fig. 2(a), is an important component in the introduced PKI Block, which consists first of a small kernel convolution to capture the current information, followed by a set of parallel deep convolutions to complement the contextual information at multiple scales and increase feature diversity. The PKI Module, a key part of the PKI Block, uses a small kernel convolution to capture current information, followed by parallel deep convolutions that complement contextual information at multiple scales, enhancing feature diversity. This is an improvement for real-time and fast capture of underwater bridge pier crack information.

Ultimately, with the efficient integration of the 1×1 convolutional layer, the model is able to delicately combine the captured local detailed features with the rich contextual features to construct fine feature representations across different channels.The 1×1 convolutional layer, as a key pivot of channel fusion, skillfully integrates the feature representations at different scales, allowing the model to comprehensively capture the fine texture of the cracks of the underwater piers along with the context of the surrounding environment. The features of underwater bridge pier cracks are accurately extracted. In this way, the PKI module is able to meticulously characterize the local features of the cracks without compromising the integrity of the local texture features.

2.3. CAA Block

To effectively capture and understand remote contextual information in underwater bridge pier crack detection, we introduce the Context Anchor Attention (CAA) Block, building upon the PKI Block shown in Fig. 2(c). The design of the CAA Block originates from the recognition of the importance of remote contextual information, and its main purpose is to capture the interdependence between different contextual regions, while enhancing the ability to perceive the central region characterization sensing capability. The CAA Block captures the global context through average pooling operation, and then refines the local region features through 1×1 convolution, and then simulates the effect of the traditional large kernel convolution by using deep-wise stripe convolution, which achieves a similar feature extraction capability with less number of parameters. In particular, the design of strip convolution optimizes the model's feature recognition and extraction of elongated shaped objects (e.g., cracks), which is crucial for the accurate detection of cracks in underwater bridge piers. In addition, the CAA module adapts to PKI modules of different depths by adjusting the size of the receptive field, where the computational kernel size of the receptive field, $k_b = 11 + 2 \times l$, is functionally related to the depth n of the PKI module. This design significantly improves the capability of UCRT-DETR in establishing long-range inter-pixel relationships, which in turn provides an efficient support method for accurate identification and analysis of underwater bridge pier cracks.

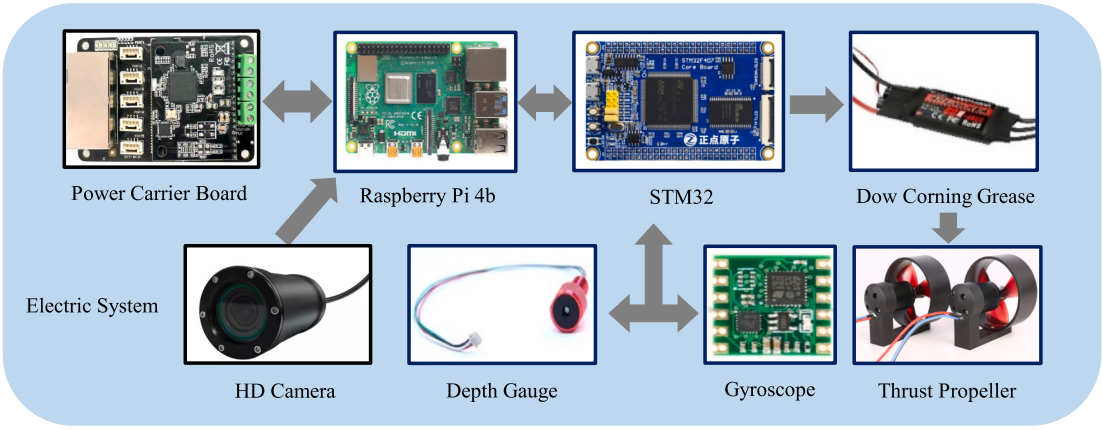

Fig. 4. Electronic bin circuit system.

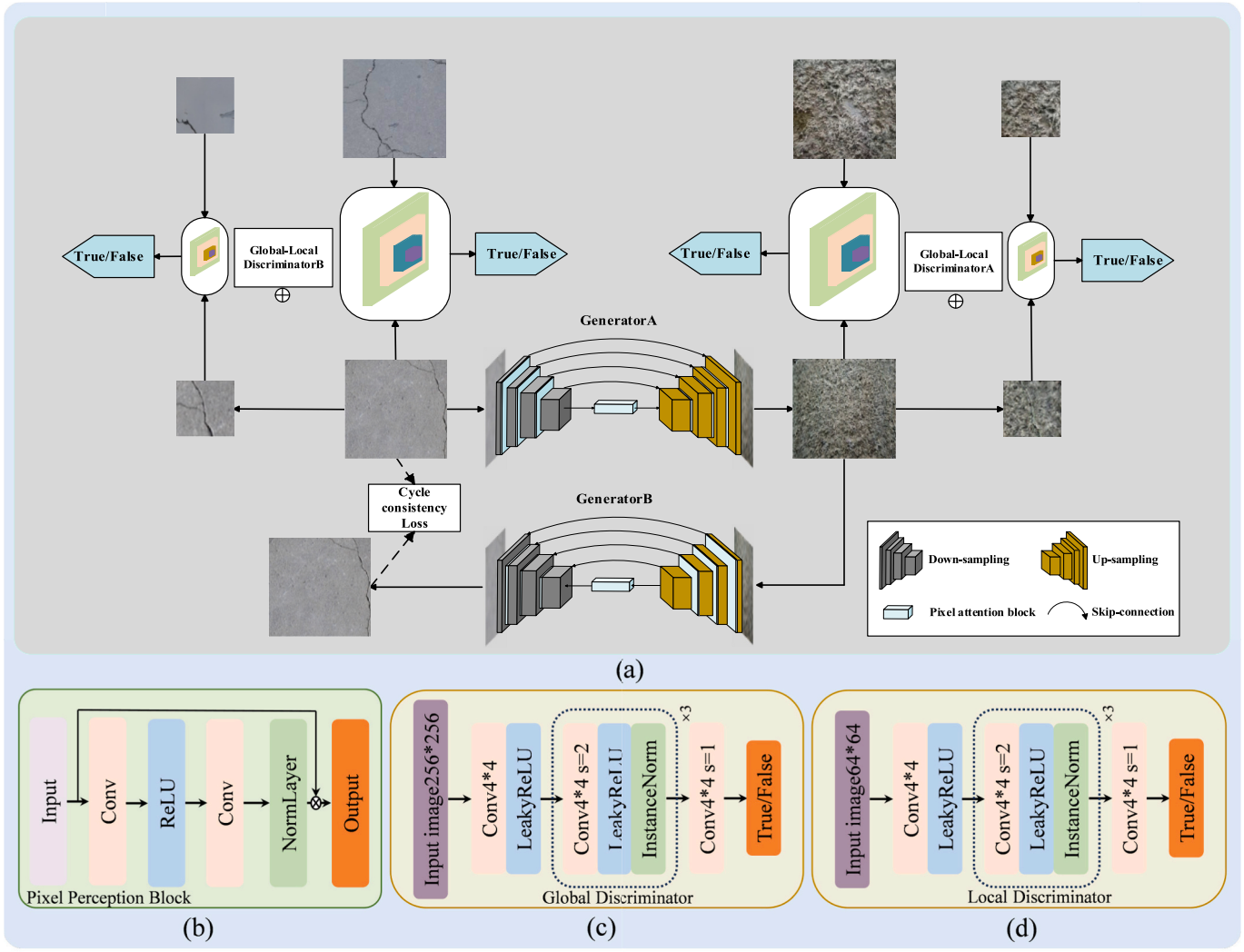

Fig. 5. Improved CycleGAN model.

3. Crack generation in underwater bridge piers

In bridge engineering, the underwater part of the pier is less, each underwater robot test collected pier wall data only a small portion of the information with cracks, there are still most of the pier wall image data, underwater pier cracks in a small number leads to poor training of the detection model, however, through the collection of multiple bridges to achieve enough crack data will lead to a significant increase in workload. To address this problem, the improved CycleGAN [36] network is considered to migrate the land bridge pier crack styles into underwater crack images, which not only expands the dataset and enhances the accuracy of underwater robotic crack detection, but also reduces the

amount of task.

3.1. Description of underwater robot hardware structure

In order to make the obtained underwater pier cracks of better quality, an ROV underwater robot was fabricated to collect underwater pier wall photographs as shown in Fig. 3.

The ROV underwater robot primarily comprises a control system, a propulsion system, a power supply system, a visual perception system, and a communication system. The STM32F407 serves as the primary controller processor for the control system, which employs sensors such as gyroscopes and depth gauges to regulate the robot's attitude, heading, and dive depth through proportional-integral-derivative (PID). The robot's mobility is powered by eight waterproof thrusters that generate five kilograms of thrust. The robot is powered by a high-power switching power supply, which is then directed to the hermetic and pressure-resistant electronic chamber to power the circuits within the chamber, as illustrated in Fig. 4. The visual perception system employs a high-definition waterproof camera as its main component to acquire structural data from the underwater wall surface of the bridge pier. The utterly automatic zoom function of the waterproof camera allows the underwater robot to rapidly and accurately capture images of the underwater pier wall while hovering at a fixed depth.

Additionally, the camera can capture clear 1080P underwater video images. The communication system's coiler utilizes a 300-m-long four-core cable (two cores for power supply and two cores for signal transmission) to supply power to the switching power supply of the underwater robot over a long distance. Simultaneously, the power carrier board in the electronic compartment converts all types of data from the underwater robot into power signals and transmits them to the communication box on the shore, which contains the power carrier board. The data is then wirelessly transmitted to the top computer PC via WiFi, making it convenient for observation and control.

3.2. CycleGAN-based defect generation

CycleGAN is a variation of Generative Adversarial Networks (GAN) [37] that uses unsupervised learning to implement image style transformation techniques. In order to learn the transformation relationship between the source and target images, it employs two Generative Adversarial Networks (GANs) to capture the stylistic features of each separately. This procedure does not necessitate any labeling information or associated image samples. Two generators operate in tandem; one converts the source image's style to the target style, while the other performs the inverse operation. To guarantee the conversion process's accuracy and consistency, CycleGAN implements cyclic constraints. The authenticity and naturalness of the generated image are guaranteed by these constraints, which ensure that the image can be restored to its original state to the greatest extent possible following the round-trip processing of style conversion. Fig. 5(a) illustrates its model structure.

Concrete fissures on land are fed into Generator A to produce cracks on underwater bridge piers. The underwater pier wall collected by the underwater robot is utilized as a style picture in domain B to conduct global and local discrimination with the generated underwater pier cracks to ensure their similarity. Simultaneously, the underwater bridge pier images are fed into generator B to produce C domain images. The co-consistency loss guarantees that the images input into generator A and the C domain images are similar. The C domain images are globally and locally discriminated from the A domain images, concluding the style migration from the A domain to the B domain.

3.2.1. Network structure

To guarantee the detection's efficacy, the generated images should be as similar as possible to the images of the pier walls collected by the underwater robot. This ensures that the color of the image remains unchanged, and only the artifactual noise of the image generated by the model is repaired. The traditional ResNet framework is replaced by the Unet network architecture in the generator's construction. The Unet [38] architecture is capable of more effectively preserving the low-level semantic information of the image during the feature extraction process, which is essential for the integrity of the reconstructed image information, in contrast to the traditional ResNet architecture. Furthermore, the Unet architecture incorporates a pixel-level attention module in each downsampling and upsampling, as illustrated in Fig. 5(b). This module allows the generator to more accurately capture and concentrate on the pixel-level details of the cracked image, thereby improving the quality of the image reconstruction. This innovative network design enables the generation of more detailed fracture images providing strong support for the subsequent detection and analysis of cracks.

The discriminator's discriminative effect will produce errors as a result of the presence of artifacts and noise in the images generated by the CycleGAN model, which has a certain impact on the detection of underwater bridge pier cracks. To mitigate this phenomenon, the global local discriminator [39] is employed to discriminate the images generated by the generator. The global-local discriminator enhances the discriminative effect on the generated images, as illustrated in Fig. 5(c), (d). Additionally, the generator gains a more precise control over the local details through the Nash equilibrium.

3.2.2. Generation of underwater bridge pier crack loss function

The quality of the underwater bridge pier crack data directly influences the effectiveness of the detection model. In the context of the image generation task, the high-quality underwater bridge pier crack image produced by the generator is subject to the discrimination of the global-local discriminator. In addition to influencing the realism and usability of the final image, the loss function of CycleGAN also determines the degree of image detail retention during the migration process. This is achieved by constraining the generator and discriminator. The loss function of CycleGAN consists of three main components, adversarial loss, cycle consistency loss and identity loss.

1) **Adversarial Loss**

Make it difficult for the discriminator to differentiate between the real and fake by generating crack images that are visually indistinguishable from real crack images. This is especially crucial for style migration, as it is not only necessary to create the cracks' style but also to guarantee that they appear realistic and authentic on an image of a submerged bridge pier. The following is the formula for training the network using least squares loss:

$$L_{GAN}^{Global}\left(G, D_Y^{Global}, x, y\right) = \left(D_Y^{Global}(G(x)) - 1\right)^2 + \left(D_Y^{Global}(y) - 1\right)^2 + \left(D_Y^{Global}(G(x))\right)^2 \tag{1}$$

$$L_{GAN}^{Global}\left(F, D_X^{Global}, x, y\right) = \left(D_X^{Global}(F(y)) - 1\right)^2 + \left(D_X^{Global}(x) - 1\right)^2 + \left(D_X^{Global}(F(y))\right)^2 \tag{2}$$

$$L_{GAN}^{Local}\left(F, D_Y^{Local}, x, y\right) = \left(D_Y^{Local}(G(x_{local})) - 1\right)^2 + \left(D_Y^{Local}(y_{local}) - 1\right)^2 + \left(D_Y^{Local}(G(x_{local}))\right)^2 \tag{3}$$

$$L_{GAN}^{Local}\left(F, D_X^{Local}, x, y\right) = \left(D_X^{Local}(F(y_{local})) - 1\right)^2 + \left(D_X^{Local}(x_{local}) - 1\right)^2 + \left(D_X^{Local}(F(y_{local}))\right)^2 \tag{4}$$

$$L_{GAN} = L_{GAN}^{Global}\left(G, D_Y^{Global}, x, y\right) + L_{GAN}^{Global}\left(F, D_X^{Global}, x, y\right) + L_{GAN}^{Local}\left(F, D_Y^{Local}, x, y\right) + L_{GAN}^{Local}\left(F, D_X^{Local}, x, y\right) \tag{5}$$

G and F are generators that are capable of converting land fractures into underwater bridge pier crack images and underwater bridge pier crack images into land crack images. and are discriminators that are

Fig. 6. Underwater robot collecting data on the wall surface of Huanghuayuan Bridge pier. (a) Huanghuayuan Bridge, (b) localized situation of the bridge pier, (c) data collected by the underwater robot.

used to make pixel-aware judgments of the generated underwater pier images of varying sizes and the corresponding visible overwater pier wall images. Similarly, and are employed to make pixel determinations on the generated global-sized and local-sized land crack images, as well as the corresponding land crack images.

2) **Cycle Consistency Loss**

The structural information of the original underwater structures is preserved by ensuring that the cyclic transformation of the image between the source and target domains remains consistent. This means that the original features of the underwater pier, such as its shape, texture, and distribution of cracks, are still plainly visible, even if the image is converted to that of a terrestrial concrete crack, as in Eq. 6.

$$L_{\mathrm{cyc}}(G,F) = E_{x\sim p_{data(x)}}[\|F(G(x)) - x_1\|] + E_{y\sim p_{data(y)}}[\|G(F(y)) - y_1\|] \quad (6)$$

3) **Identity Loss**

When dealing with images that already possess the desired stylistic features in the target domain, identity loss is of particular importance. It ensures that the generator does not over-modify these images, thereby preserving the details of the fractures and preventing the loss of critical visual information during the style migration process.

$$L_{idt} = E_{y\sim p_{data}(y)}\left[\,|F(x) - x\|_1\right] \quad (7)$$

The loss function's objectives, as previously described, are combined to ensure that the generator generates images that prioritize visual realism, structural consistency, and style preservation. This provides the generator a balanced optimization objective to generate images that are both reminiscent of underwater piers and resemble the style of land-based concrete fractures. It is illustrated in Eq. 8:

$$L = \gamma_1 L_{GAN} + \gamma_2 L_{Idt} + \gamma_3 L_{cyc} \quad (8)$$

where, γ_1, γ_2, $\gamma_3$are the weights of each loss function.

4. Experiments and results

This chapter provides a systematic overview of the experimental pre-preparation phase, which encompasses the acquisition and processing of datasets, as well as the establishment of critical parameters in the experimental design. The proposed methodology is rigorously evaluated for practicality and validity by evaluating a series of quantitative metrics and combining them with ablation experiments to guarantee its reliability and feasibility in engineering practice.

4.1. Preparation of dataset for generating underwater bridge pier cracks

4.1.1. Dataset acquisition

Unpaired datasets are necessary for the CycleGAN algorithm to be implemented. Ensure that the land fissures have a more uniform background in order to more effectively migrate the land crack information while retaining the additional details of the bridge pier walls. The training set A is comprised of 2568 sheets of publicly available bridge concrete fracture data, while the test set A is composed of 555 sheets. The training and test sets are primarily composed of two sets of data. The first element is obtained from [40], in which 1568 sheets are selected as the training set and 500 sheets as the test set. The bridge fracture data is 1024*1024 pixels in size. The training and test sets for the second part are the primary dataset of bridge concrete crack images, which has a size of 200*200 pixels. This dataset is divided into two sets of data: data with cracks and data without cracks. The training set consists of 1000 sheets of crack data, while the test set consists of 55 crack data.

Dataset B is divided into two components. As illustrated in Fig. 6(a), the Huanghuayuan Bridge in Jiangbei District of Chongqing Municipality was opened to traffic in 2013. This river crossing, which connects Jiangbei District and Yuzhong District in Chongqing Municipality, China, is dependent on the underwater robot to acquire data. The bridge

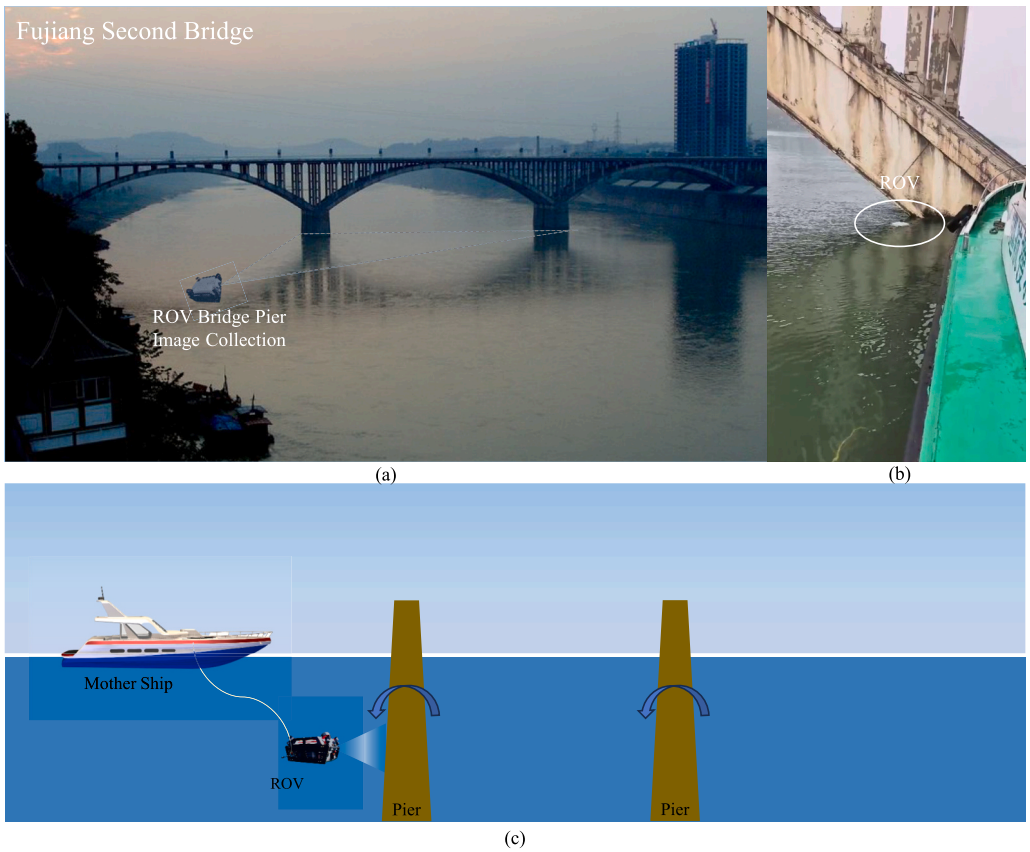

Fig. 7. Underwater robot acquisition of wall data for the piers of the Fuling River Second Bridge. (a) Acquisition location of the Fuling River Second Bridge. (b) Actual data collection by the robot (c) Simulated underwater robot operation.

Fig. 8. Cracks in terrestrial concrete piers (top), data collected from the wall of the Huanghuayuan pier, data collected from the wall of the pier of the second bridge on the Fuling River (bottom).

has a total field of 1208 m. The underwater bridge pier is susceptible to cracking as a result of its age and high traffic. The robot light is not activated as an auxiliary visual perception device, as the acquisition process occurs during the day and the light is more than adequate, as illustrated in Fig. 6(b). The underwater robot uses the dive fixed depth PID algorithm to acquire underwater pier wall information at various depths, as illustrated in Fig. 6(c).

The water discharge beneath the Huanghuayuan Bridge is slow in the

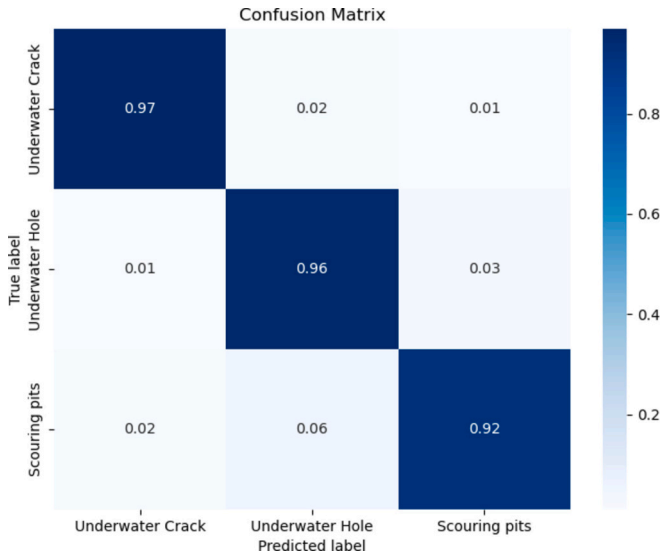

Fig. 9. Confusion matrix for classifying underwater bridge pier defects with ResNet.

Table 1
Distribution of data sets for each model.

Model	Training	Validation	Test	Total
Improved CycleGAN	A:2568 B:3029	\	A:555	6152
UCRT-DETR	2498	312	363	3123

Table 2
Hyperparameter settings for network training.

Model	Training parameters	Value	Model	Training parameters	Value
CycleGAN	Input size	2560 × 1440	RT-DETR	Input size	256 × 256
	Batch size	8		Batch size	16
	Epochs	200		Epochs	200
	Optimiser	Adam		Optimiser	AdamW
	Momentum	0.9		Momentum	0.9
	Learning rate	0.0001		Learning rate	0.0001

dry season. The underwater robot is capable of traversing the water at a speed of 0.3 m per second, scanning to capture video, and processing the video every 2 s to intercept a path to acquire 2560 * 1440 pixel images. In total, the robot will collect data from 1203 piers. The water scour, the bridge bearing, and the pier wall images are all distinct despite the fact that diverse underwater concrete walls are roughly similar. This is done

Table 3
Assessment of the quality of the images generated by the model for different number of datasets (best scores are in bold).

Data A	Data B	PSNR	SSIM	UIQM	UCIQE
1000	1000	19.3245	0.4596	0.6393	0.2742
	1500	19.2034	0.4846	0.6294	0.2656
1500	1000	18.4026	0.4495	0.6038	0.2543
	1500	20.1659	0.4938	0.6992	0.3059
	2000	20.7953	0.5230	0.7239	0.3210
2000	1500	19.9384	0.5187	0.7184	0.3194
	2000	21.0319	0.5492	0.7321	0.3319
	2500	20.8475	0.5389	0.7285	0.3214
2500	2000	20.4684	0.5194	0.7089	0.3001
	2500	21.2258	0.5932	0.7599	0.3595
	3000	**21.3746**	**0.6257**	**0.7783**	**0.3730**
3000	2500	20.8353	0.5758	0.7294	0.3267
	3000	20.4938	0.5693	0.7234	0.3195

Table 4
Evaluation of model-generated image metrics with different hyperparameters (best scores in bold).

γ_3	PSNR	SSIM	UIQM	UCIQE
5	19.2498	0.5907	0.7239	0.3284
10	**21.4356**	**0.6323**	**0.7823**	**0.3801**
15	20.8347	0.6139	0.7697	0.3672
20	20.6872	0.6137	0.7583	0.3613

to prevent the pier wall information from being too similar, which could result in overfitting in the model training. Therefore, the second bridge of the Fujiang River, which is situated in Hechuan City, Chongqing City, and spans the Fujiang River, was completed and opened to traffic in 1996. Hechuan City is the sole automobile access point to the city, and the traffic flow is substantial. The underwater pier image data is depicted in Fig. 7. In this regard, acquiring images of the underwater bridge pier is more appropriate.The river current is sluggish, at approximately 0.2 m per second, when the robot power supply is powered by the mother ship. The bridge pier wall is scanned by the underwater robot using a fixed depth PID and attitude PID. The video was processed by intercepting one photo per second, resulting in the acquisition of 1826 image data of 2560*1440 pixels, a total of 3029 bridge pier wall data. Fig. 8 illustrates dataset A and dataset B. The dataset of the detection model is created by combining the generated fracture data and the resultant data after testing in Test Set A. This dataset is then divided into the training set, validation set, and test set in a ratio of 8:1:1. In the interim, the test set is supplemented with 53 sheets of actual underwater bridge pier crack data collected by the underwater robot to validate the model's performance.

4.1.2. Multiple pier defect data classification

To more comprehensively evaluate the performance of the UCRT-DETR model, it is tested not only on the underwater bridge pier crack data generated by the improved CycleGAN, so a small amount of real underwater bridge pier crack data collected by the underwater robot is used to verify the performance of the test and inspection model, however, the underwater robot obtains multiple categories of defect data (cracks, holes, scouring pits, etc.) during the inspection of the bridge piers. So a classification model is used to categorize the multi-category defect data.

Underwater bridge pier crack data is difficult to distinguish by the interference of the water body, and there are large differences between the land cracks. However, through the comparison of other underwater bridge pier defect data collected by the underwater robot, it was found that the same category of defects with the land data is similar, not like the underwater bridge pier crack data is difficult to distinguish between the defects and the background there is a big difference, it is easier to identify. Therefore, the defect data of other categories of land were utilized as the dataset [41] for the classification model to train the classification model. A total of 700 sheets of each of the 3 categories of defect data. It was randomly partitioned into training set, validation set and test set in the ratio of 8:1:1.

The ResNet model has a strong feature extraction capability and a strong ability to distinguish between different kinds of image data, so the ResNet model is chosen as the classification model to distinguish between several kinds of underwater bridge pier defects. The hyperparameters of the ResNet model are fine-tuned to obtain the best defect classification performance. The data collected by the underwater robot is used as a test set for the ResNet model, The training results are shown in the confusion matrix in Fig. 9. When classifying several kinds of underwater bridge pier defect data, it can be seen that ResNet can effectively classify three different categories of defect data, and its classification accuracy of ResNet is all up to 95 % or more. This proves that the trained ResNet classification model works better. There are 56 sheets of crack data, 71 sheets of hole data, and 8 sheets of washout pit

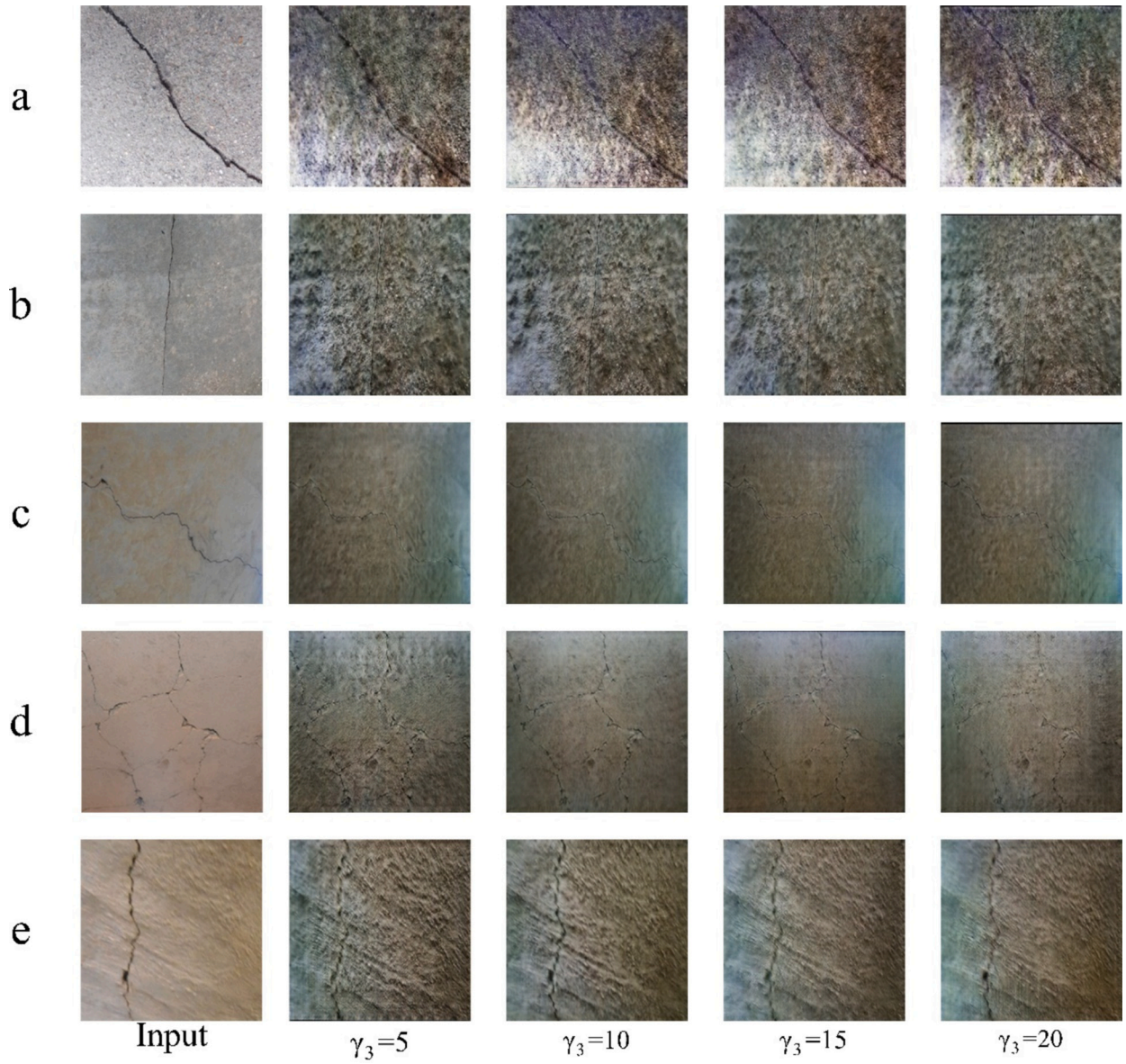

Fig. 10. Images generated by improved CycleGAN with different hyperparameters.

data. The amount of defect data is small, from which about 53 sheets of data containing only cracks can be filtered as a test set to test the performance of the UCRT-DETR model. Table 1 illustrates the allocations of the two model datasets.

4.1.3. Labeling of datasets

All crack images in the dataset must be annotated prior to the use of UCRT-DETR for fracture recognition. The open source online annotation application MakeSense AI [42] was employed for this task. It offers a user-friendly and intuitive interface that enables users to rapidly annotate images and begin using the tool, in contrast to other options. Users are not required to install or configure Makesense AI on their local computers, which simplifies the process of using it, as it is an online application. It is compatible with a variety of operating systems. The tool's flexibility is enhanced by its support for a diverse array of annotation types, such as rectangles, points, lines, and polygons. The annotation data can be exported to a variety of commonly used formats, including YOLO, VOC XML, VGG JSON, and CSV, as required. This facilitates compatibility with various machine learning models.

4.2. Experimental configuration

The experimental platform is running on a computer that is configured with a 15vCPU Intel(R) Xeon(R) Platinum 8358P CPU@2.60GHz and an NVIDIA A40 (48G) GPU, and is running the Ubuntu20.04 operating system. The Pytorch 1.11.0 framework has been completed to generate the underwater bridge pier crack generation model and detect the bridge pier fractures for this experiment. The selection of multiple training parameters, such as the size of the input images, epochs, batch size, learning rate, and optimization algorithm significantly influences the efficacy of deep learning models. The model training in this investigation was conducted using the training parameters listed in Table 2.

4.3. Evaluation of Improved CycleGAN Results

4.3.1. Evaluation indicators

This section conducts quantitative experiments and qualitative analyses. Underwater bridge pier fractures are produced by employing unpaired datasets. The subjective evaluation results are supplemented by the qualitative experiments. Furthermore, quantitative experiments

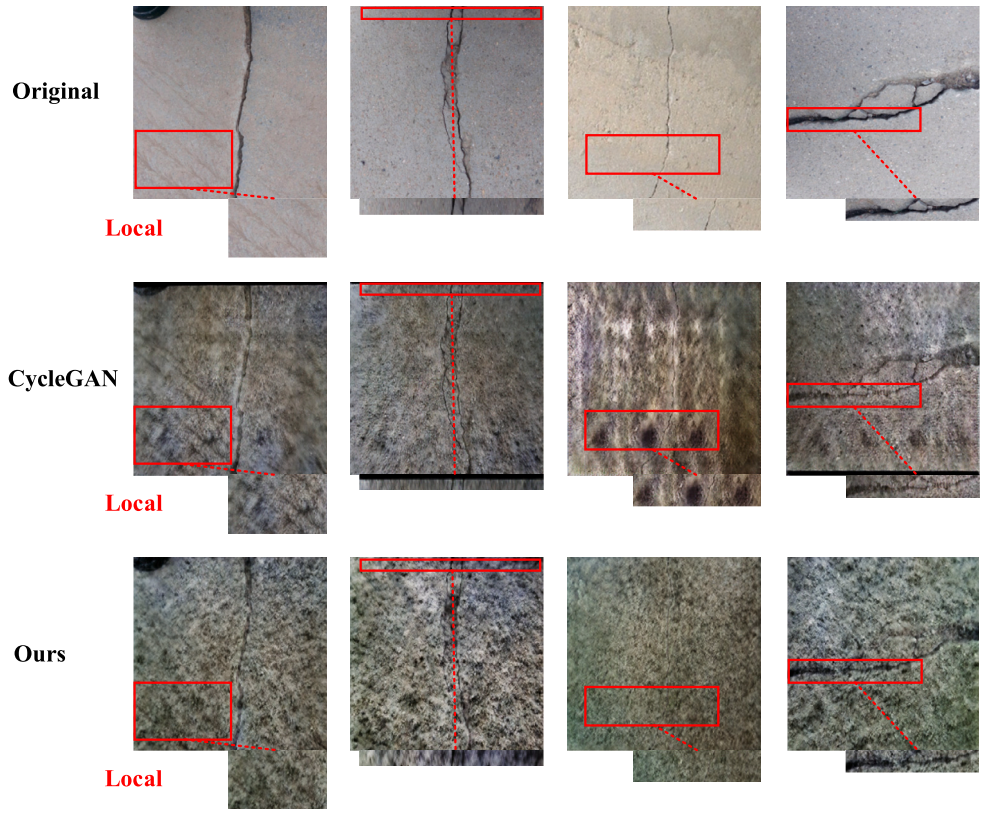

Fig. 11. Comparison of underwater bridge pier crack generation (CycleGAN and improved algorithms, local differences are shown in red). (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

Table 5
Comparison of performance metrics evaluated with the original model (black bolding indicates best scores).

	PSNR	SSIM	UIQM	UCIQE
CycleGAN	17.8943	0.5687	0.7459	0.3445
Ours	**21.4356**	**0.6323**	**0.7823**	**0.3801**

are the primary metrics for presenting the evaluation results in an objective manner. The Structure Similarity Index Measure(SSIM) [43] was employed to determine the quality of underwater bridge pier fractures, with higher scores indicating better quality. The Peak Signal-to-Noise Ratio (PSNR) was also considered.

The performance of the underwater bridge pier crack image is evaluated using Underwater Image Quality Measures(UIQM) [44], Underwater Color Image Quality Evaluation Metrics (UCIQE) [45], and UIQM in conjunction with underwater image colorfulness measure (UICM), underwater image sharpness measure(UISM), and underwater image contrast measure(UIConM), as illustrated below. This is due to the fact that the image occurs within the underwater scene.

$$UIQM = c_1 \times UICM + c_2 \times UISM + c_3 \times UIConM \tag{9}$$

$c_1 - c_3$is the parameter used to balance the three preferences.

$$UCIQE = \gamma_1 \times \sigma_c + \gamma_2 \times con_l + \gamma_3 \times \mu_s \tag{10}$$

UCIQE is a linear combination of color concentration, saturation, and contrast that is employed to quantitatively assess the non-uniform color cast, distortion, and low contrast of underwater images. The image quality evaluation index without the reference image is depicted in Eq.

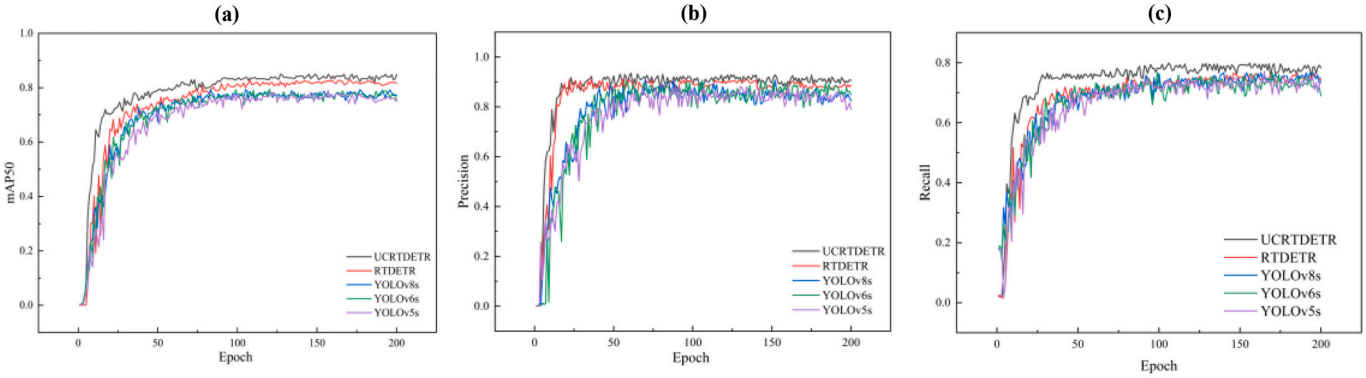

Fig. 12. Comparison of Precision, Recall, mAP50 during training of the proposed UCRT-DETR algorithm, RT-DETR algorithm, YOLOv5s algorithm, YOLOv6s algorithm, YOLOv8s algorithm, (a) mAP50, (b) Precision, and (c) Recall.

Table 6
Comparison with other advanced target detection algorithms.

Model	P	R	mAP@0.5	mAP@0.5–0.95	F1	FPS	FLOPS
YOLOv5s	0.87	0.74	0.77	0.56	0.79	68.96	23.8
YOLOv6s	0.89	0.76	0.79	0.58	0.73	84.89	44.0
YOLOv8s	0.90	0.76	0.78	0.59	0.82	138.89	28.4
RT-DETR	0.90	0.76	0.82	0.61	0.78	97.03	56.9
UCRT-DETR	0.92	0.80	0.84	0.66	0.85	87.47	57.7

10. Here, σ_c represents the standard deviation of chromaticity, con_l represents the luminance contrast, μ_s represents the mean saturation value, and $\gamma_1 - \gamma_3$ represents the weighting factor.

4.3.2. Validation of the number of datasets

Experimental validation was carried out to determine the effect of the number of data sets on the quality of the generated underwater bridge pier cracks. Huang et al. [46] used an AUV to take 984 crack data of the dam and regarded it as dataset A. Meanwhile, an underwater robot was used to take 678 data and set it as dataset B. Ye et al. [30] used 2000 land defect datasets as the training set A for the CycleGAN model. Additionally, 2000 underwater datasets were used as the training set B for the CycleGAN model to realize the generation of diseased image data of the underwater buildings. Considering the consumption of computational resources as well as too little data leads to poor results of the improved CycleGAN generation. 1000,1500,2000,2500,3000 pieces of data are selected for training in each of the dataset A and dataset B. The experimental configurations and hyper-parameters used each time are the same, and after testing experiments to verify that when the difference between dataset A and dataset B is about 1000 sheets of hyper-parameters, the quality of the generated images is verified to be worse and the visual perception effect is not good after the four evaluation indexes. Therefore, here exclude the two data difference of about 1000 cases, the experimental results are shown in Table 3.

From Table 3, it is easy to find that when the land crack data reaches about 2500, and when the underwater bridge pier wall image reaches about 3000, all the four indexes reach the best, which proves that the model generates the best quality of data, and thus verifies the reasonableness of the selection of the improved CycleGAN dataset.

4.4. Loss function optimal hyperparameters

The hyperparametersγ_1, γ_2, γ_3 have an important effect on the quality of the images generated by the improved CycleGAN, so it is important to choose the correct hyperparameters of the loss function, which are inspired by [47,48] inspired, usually the choices of $\gamma_1 = 1$, $\gamma_2 = 10$, and γ_3 are indeterminate, and the ranges are generally between 5 and 20, so four different values of γ_3 (5, 10,15,20) are used to train the improved CycleGAN, as shown in Fig. 9 and Table 4. The appropriate value of γ_3 is determined by analyzing and comparing the quality of the generated underwater bridge pier cracks.

Different choices of hyperparameters make the image quality generated by the improved CycleGAN different, from Table 4 and Fig. 10, it can be found that when γ_3=5, the PSNR metric parameter, SSIM metric parameter, UIQM metric parameter, and UCIQE metric parameter are all poor, and from Fig. 10(a) and d, it is found that there are color deviation and artifacts in the image; when γ_3=15 and γ_3=20, the four evaluation metrics are improved, but still not optimal, as can be seen from the figure, compared to γ_3=10, there are color deviations in Fig. 10(a), and blurring in Fig. 10(b), (e). Therefore, the subsequent experiments are based on generating underwater bridge pier crack images from the training model with γ_3 set to 10.

4.5. Improved qualitative evaluation of CycleGAN

The same training sets A and B are employed to experimentally train the CycleGAN and the improved model. The batch size is set to 8, and 200 epochs of training are conducted, with the same optimizer and parameter configurations. The proposed method and the CycleGAN model are used to compare the images of underwater bridge pier cracks in test set A. Fig. 11 illustrates the random selection of numerous underwater bridge pier fractures for subjective visual analysis. The enhanced algorithm achieves the most favorable outcomes by mitigating the effects of fogging, clarity enhancement, and color distortion, as demonstrated by the CycleGAN comparison test. The initial image processed by CycleGAN exhibits clear fogging and inadequate migration of terrestrial concrete fissures, rendering the subsequent training process challenging due to the inability to differentiate the cracks' characteristics accurately The final three images exhibit the same issue: the generated fissures exhibit poor local characteristics, the color is distorted, the overall visual perception is purple, and the images contain artifacts.

4.6. Improved CycleGAN quantitative evaluation

The proposed method's efficacy is quantitatively assessed using four metrics for the test set A, as illustrated in Table 5. The same hyper-parameters as well as experimental configurations were selected to start the comparison experiment. The enhanced algorithm exhibits the highest scores in each metric when contrasted with the original algorithm. This demonstrates the model's reliability in generating fracture data.

4.7. UCRT-DETR assessment indicators

The primary evaluation metrics of the model are Precision (P), Recall (R), mean average precision(mAP), and F1 score. These metrics are employed to assess target detection. The proportion of correctly predicted samples among all detected targets and the proportion of correctly predicted samples among all targets are denoted by P and R, respectively. The F1 score is the reconciled mean of Precision and Recall, while mAP is the average area under the Precision-Recall curve for all categories. Eqs. (11)–(14) can be used to calculate it, where k represents the number of detection categories, TP, FP, TN, and FN represent true positive, false positive, true negative, and false negative, respectively, and the category value in this investigation is 1. Furthermore, frames per second (FPS) is a critical metric for assessing the model's ability to detect in real time.The model's capacity to analyze frames per second is represented by FPS. Accuracy and recall are equally critical in the detection of underwater pier cracks, as misclassification or false negatives may have consequences for the safety and maintenance of underwater piers. Consequently, this investigation concentrates on the model's overall performance, with the F1 score, mAP@50, mAP@50–95, and FPS serving as the most critical evaluation indices.

$$Precision = \frac{TP}{TP + FP} \tag{11}$$

$$Recall = \frac{TP}{TP + FN} \tag{12}$$

$$F1 = \frac{2 \times P \times R}{P + R} \tag{13}$$

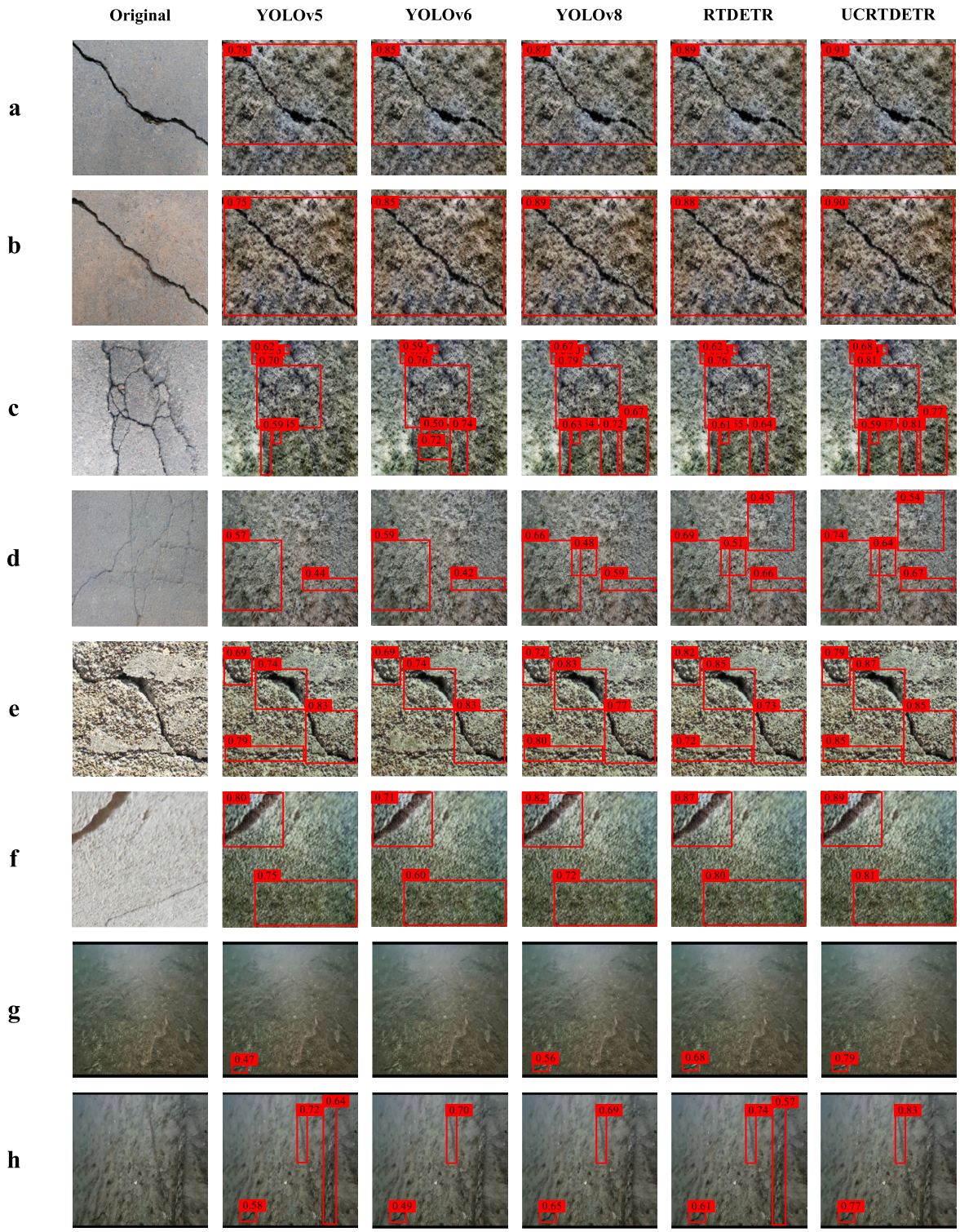

Fig. 13. Comparison of detection results for different types of cracks: (a) conventional cracks; (b) conventional cracks with yellow background; (c) reticulated cracks; (d) fine reticulated cracks; (e), (f) mixed wide-body and elongated cracks; (g), and (h) real underwater pier cracks. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

$$mAP = \frac{1}{k}\int_0^1 Pre(Rec)dRec \tag{14}$$

4.8. Experimental comparison of this method with other state-of-the-art methods

In order to confirm the validity and reliability of the improved model and the reasonableness of the RT-DETR model as a baseline model for detecting underwater bridge pier cracks, comparisons were conducted with a variety of state-of-the-art algorithms, including YOLOv5s, YOLOv6s [49], YOLOv8s, and RT-DETR, using the same hyper-parameter settings in all models. Fig. 12 illustrates the training outcomes. The mAP curves, accuracy, and recall of multiple models during training are illustrated in Fig. 12 (a)-(c). It demonstrates that UCRT-

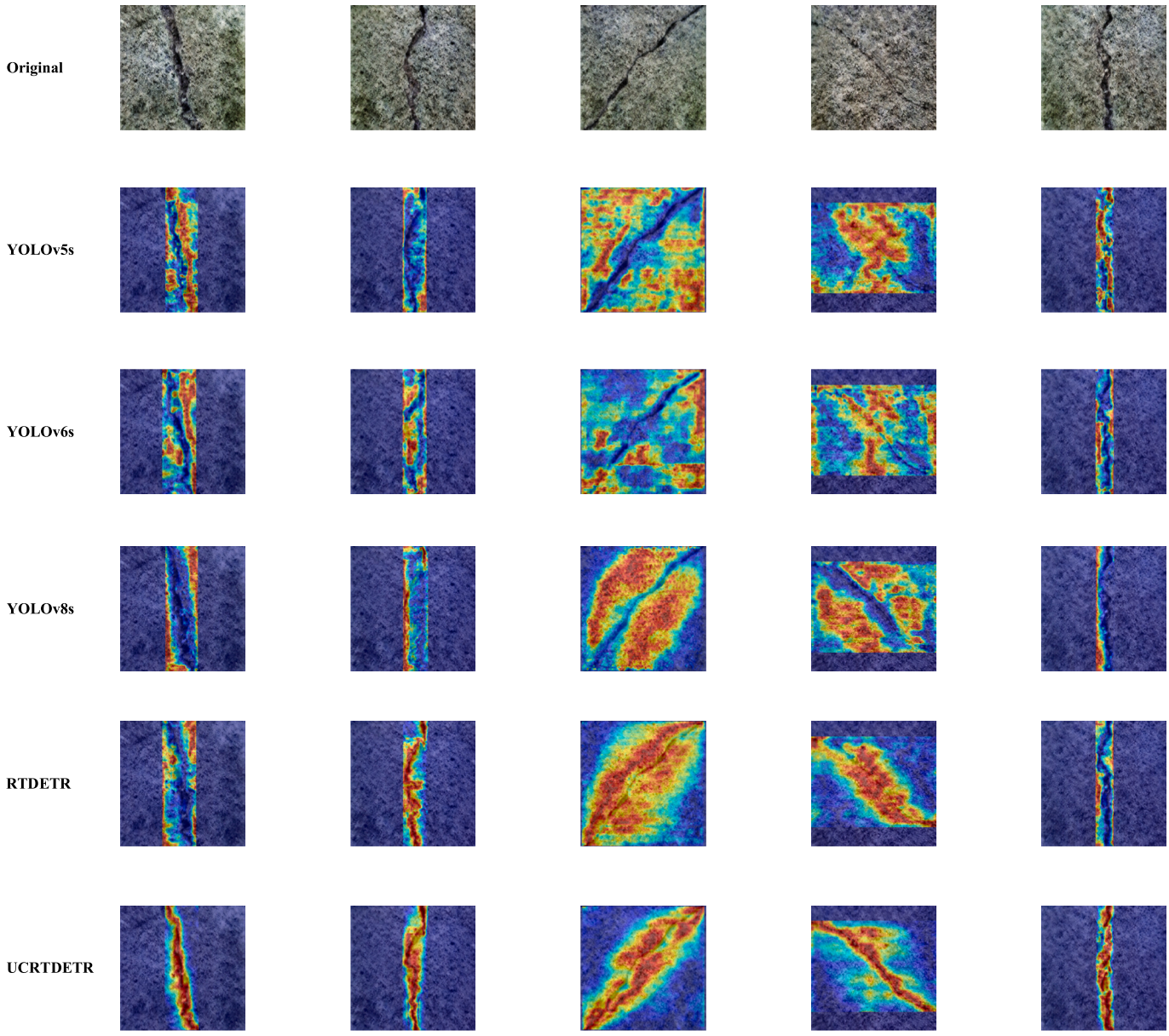

Fig. 14. Comparison of Grad-Cam plots of different algorithms.

Table 7
Results of network ablation experiments.

Model	P	R	mAP@0.5	mAP@0.5–0.95	F1	FPS
RT-DETR	0.90	0.76	0.82	0.61	0.78	97.03
w/o PKI Module	0.90	0.78	0.82	0.62	0.83	102.32
w/o CAA Block	0.91	0.79	0.83	0.65	0.84	90.58
UCRT-DETR	0.92	0.80	0.84	0.66	0.85	87.47

DETR outperforms numerous other methods with superior evaluation indices. It demonstrates that the UCRT-DETR model is appropriate for the detection of cracks in underwater bridge piers. In contrast, Table 6 displays additional index scores, which demonstrate that RT-DETR outperforms the YOLO series of methods in numerous indexes, particularly mAP, which reaches 0.82. Additionally, RT-DETR demonstrates a clear advantage in comprehensive performance when contrasted with the YOLOv5s and YOLOv6s methods. For instance, RT-DETR's accuracy is 3 % to 90 % higher than that of YOLOv5s, and the F1 score is also 3 % higher. Conversely, the F1 score of YOLOv6s algorithm is markedly higher by 9 %. Furthermore, it is determined that RT-DETR surpasses the YOLOv5s and YOLOv6s methods in terms of detection speed through the comparison of FPS metrics. It is important to mention that YOLOv8s is faster than RT-DETR in terms of detection speed. However, it also meets the requirement for real-time detection of fractures in underwater bridge piers with the same F1 score. Additionally, the average accuracy index indicates that RT-DETR surpasses the YOLOv8s method by 4 %. The RT-DETR is a reasonable fundamental underwater bridge pier crack detection model, as evidenced by the aforementioned evaluation indexes.

To further assess the validity and reliability of the improvement and the reasonableness of the UCRT-DETR model for detecting underwater pier fractures, a test was conducted on the underwater pier crack data generated by the improved CycleGAN. The detection outcomes of the five methods in this test set are compared in Fig. 13. UCRT-DETR has the maximum recognition accuracy rate, as demonstrated in Fig. 13(a), (b), when bridge images with varying background colors are used as the test

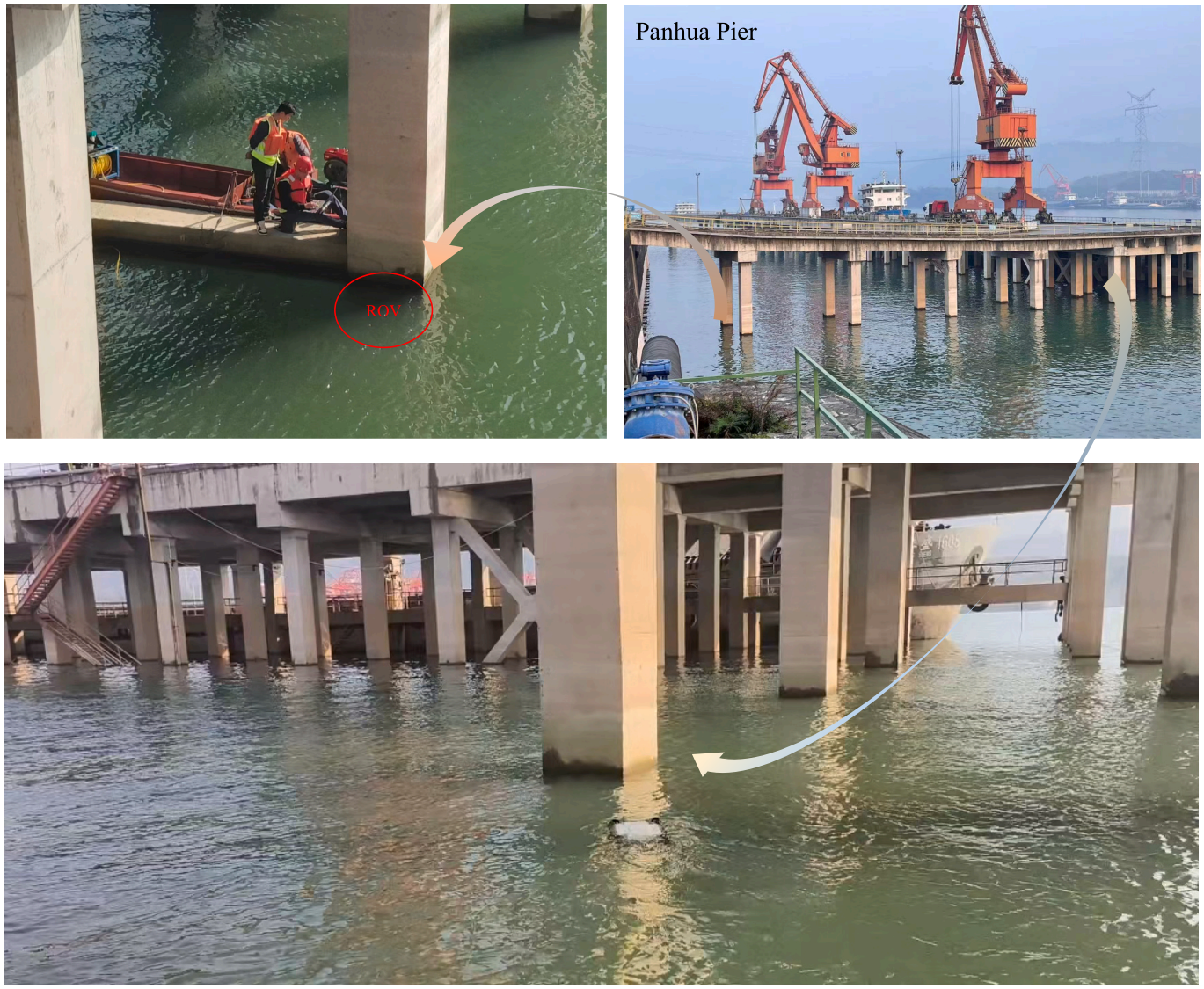

Fig. 15. ROV actual field test.

data. All methods are capable of accurately identifying cracks in simple images. YOLOv5s, YOLOv6s, and RT-DETR all failed to detect the reticulation cracks, as illustrated in Fig. 13(c). Additionally, YOLOv6s inaccurately identified the cracks. The recognition accuracy of the three methods is subpar. While both YOLOv8s and UCRT-DETR are capable of detecting underwater bridge pier fractures, UCRT-DETR is more precise. It is not difficult to determine that several YOLO detection methods have overlooked detection for slender cracks, as illustrated in Fig. 13(d). Slender fractures are accurately detected by RT-DETR and UCRT-DETR. In order to confirm the detection effect of each model on mixed cracks, as illustrated in Fig. 13(e), (f), which includes elongated fissures and wide-body cracks. The majority of the models are capable of identifying the fractures, and the comparison reveals that UCRT-DETR has the most effective detection effect for YOLOv6s with leakage detection in (e). To evaluate the efficacy of the enhanced model in identifying genuine fractures. The underwater robot's actual data on underwater bridge pier cracks is also assessed as a test set. Two of them are sampled for visual evaluation comparison, as illustrated in Fig. 13(g), (h). It is evident that the fine fractures YOLOv6s are experiencing a leakage detection situation. There is an incorrect detection circumstance for YOLOv5s and RT-DETR, which misidentifies the twine as an underwater bridge pier crack. UCRT-DETR is the most effective in detecting fractures among the two images. The RT-DETR model is not only enhanced in terms of detection efficacy, but it also has a significant engineering practical value, as a result of the comprehensive improvement.

Heat maps of YOLOv5s, YOLOv6s, YOLOv8s, RT-DETR, and UCRT-DETR were generated in this investigation using (Grad-CAM++) [50]. The Grad-CAM++ backpropagation model's output, which is the gradient, is denoted by blue lower values and red larger values. This method comprehensively and actively represents the model's region of interest. The results of the investigation are illustrated in Fig. 14. UCRT-DETR, as illustrated in Fig. 14, prioritizes the crack's center point over numerous other techniques for both tiny and large-scale cracks. Other methods concentrate on the crack's perimeter, are more susceptible to the interference of external noises, and are less sensitive to the crack's center. Additionally, their features are more cluttered. Consequently, it is logical to select UCRT-DETR as the method for detecting the fractures in underwater bridge piers.

5. Ablation studies

This section conducts a series of ablation experiments to evaluate the validity of each improvement point and their combined effects in order to confirm the improvement effect of the proposed UCRT-DETR network

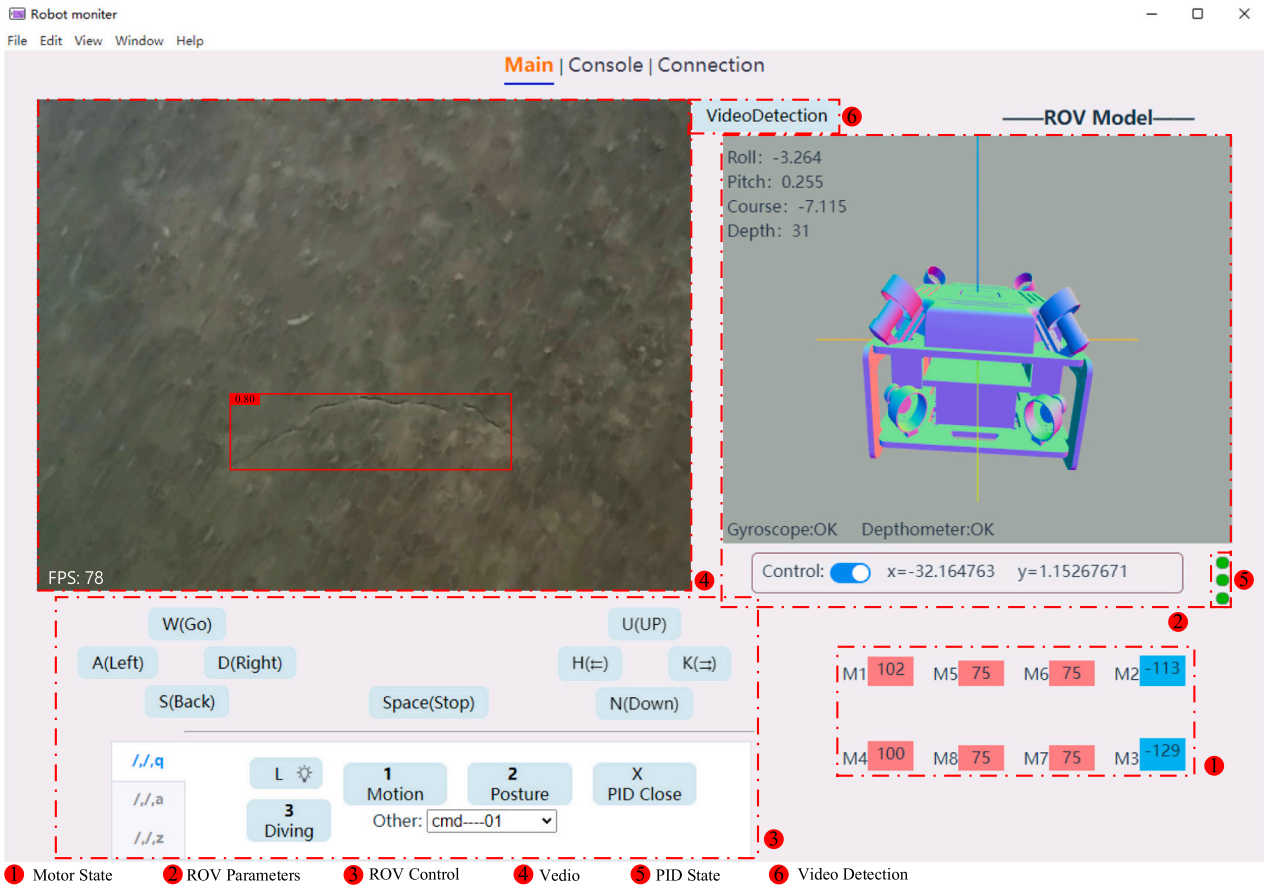

Fig. 16. Underwater robot upper unit visualization interface.

framework in comparison to the RT-DETR network. The results of the investigations are illustrated in Table 7.

The RT-DETR and improved results are illustrated in Table 7. It is primarily divided into evaluation tests following the modification of the RT-DETR backbone and the addition of the PKI Module and CAA Block. The Recall has been enhanced, the mAP and precision rate have remained unaffected, and the FPS has been enhanced as a result of the backbone change. This suggests that the YOLOv8 backbone has the capacity to fuse information from feature maps at various levels, thereby enhancing the model's capacity to detect targets at various scales. The FPS decreases by 6.6 %, despite the fact that the majority of the metrics are enhanced following the replacement of the C2f Module with the PKI Module. The PKI Module's capacity to characterize the local features of fractures in detail and enhance performance is demonstrated. The F1 score is enhanced by 7 % in comparison to RT-DETR, and the mAP@50 is enhanced by 2 % with the introduction of the CAA Block into the PKI Block.

In summary, the UCRT-DETR that was proposed in this study offers a substantial enhancement in the detection of underwater bridge pier cracks. In comparison to RT-DETR, the mAP increased by 2 %. Nevertheless, the FPS of the enhanced network model is lower than that of the RT-DETR model, which is only 87.47. Despite this, the detection speed is still sufficient to satisfy the requirements of underwater robot inspection. Consequently, this network model that has been proposed has the potential to substantially improve the capacity to identify cracks in the design framework while simultaneously ensuring real-time performance.

6. Practical applications

At Panhua Pier in Chongqing Municipality, there are numerous mounds of poured concrete that are approximately 15 m in length. The impact of the piles on the bridge piers, which was the result of the crew's operational errors and the vehicles that transported the piles over the years, caused cracks in the piles. Underwater autonomous high-definition cameras captured the video stream, which was then transmitted to an onshore PC via a buoyant cable for communication, as shown in Fig. 15.

As illustrated in Fig. 16, the practical application of the UCRT-DETR model in engineering is evaluated by constructing an upper computer visualization interface that allows for the real-time observation of the fracture effect of the underwater robot detecting the pile foundation. The ROV crack detection system software is developed using Visual Studio Code 2019 and consists of a human-computer interaction in the user interface, real-time parameter configuration and data transmission, and online upgrade features. It is designed with the JavaScript language and the VUE framework, and is packaged into executable files through electrons. The software is known for its stable and simple operation, as well as real-time parameter configuration and data transmission, as well as online upgrade functions. Additionally, it facilitates human-machine interaction. The user is only required to select the corresponding function buttons to realize the functions they require after the software is opened. This operation completes the relevant motion operation of the underwater robot and occupies minimal memory. Simultaneously, the software is developed using the object-oriented programming method, which is extremely portable and capable of accommodating the expansion of functions. The interface is equipped with a number of buttons, each of which corresponds to a unique encoded code, in order to accomplish the precise control and detection capabilities of the underwater robot. As shown in Fig. 17, The interface displays the image information sensed by the underwater robot's camera and visualizes the underwater bridge pier fracture image when it is detected. The upper

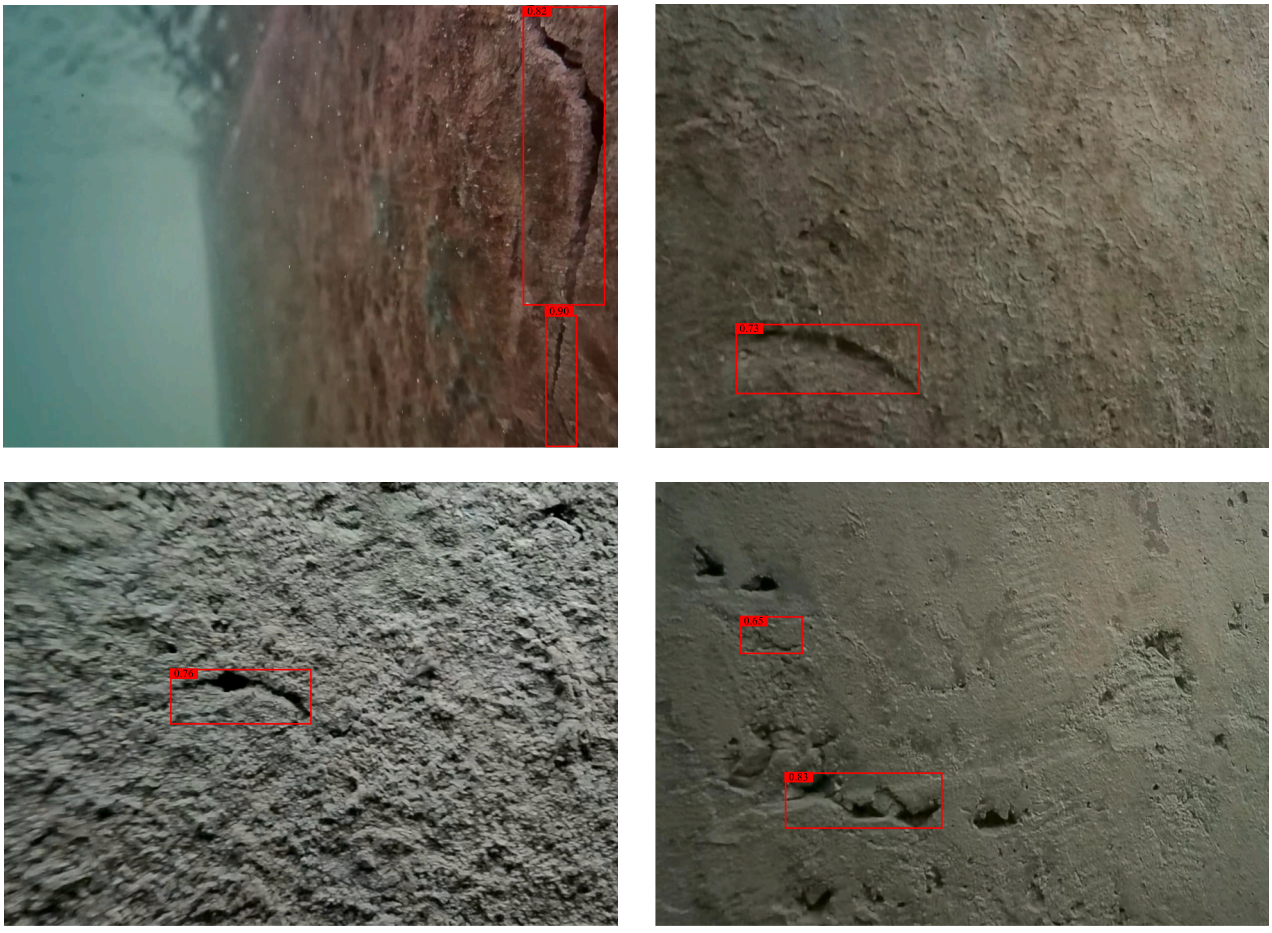

Fig. 17. Underwater cracks detected by ROV.

computer software also adds a video streaming FPS display function to reflect the real-time nature of the underwater robot when detecting underwater bridge pier cracks. The results of the detected underwater crack images as well as the accuracy information are saved locally on the computer, with an average detection accuracy of 80 %. Additionally, it furnishes real-time data on the precise location of underwater pile fracture defects. The depth information of the precise underwater crack defect image is displayed through Depth, which is the distance from the underwater pier fracture to the water surface. This is implemented to simplify the subsequent restoration job for the maintenance personnel.

7. Conclusion

Crack defects in underwater bridge piers are unavoidable, as they are a critical and vulnerable component of bridges. In order to address the issue of insufficient crack data for underwater bridge piers, a global local discriminator and pixel attention mechanism are introduced. Additionally, a crack generation method for underwater bridge piers is proposed, which is based on an improved CycleGAN. This method replaces the generator of CycleGAN with the U-Net framework and incorporates the pixel attention mechanism after each downsampling and upsampling. This method is capable of generating finer crack images and simultaneously replaces the discriminator with a global local discriminator to mitigate the artifacts and noise produced by the generator. The improved CycleGAN algorithm surpasses the original model by 16.52 %, 10.06 %, 4.65 %, and 9.37 % when comparing the PSNR, SSIM, UIQM, and UCIQE. It offers robust support for the subsequent detection and analysis of cracks.

The UCRT-DETR network is proposed to address the issue of low recognition rates of cracks identified by conventional detection models as a result of the complex water body environment. This network replaces the RT-DETR backbone network with the YOLOv8 backbone, which improves the model's ability to extract image features. Additionally, the C2f has been replaced with the PKI Block, which consists of the PKI Module and CAA Block, and integrates the various scales of cracks. To enhance the detection accuracy, the fissures' subtle texture and the background of the surrounding environment are comprehensively captured through the integration of feature representations at various scales. The model recommended in this paper is more effective in capturing the crack features of underwater bridge piers, particularly when compared to the current popular detection algorithms. This is particularly true for the issue of small differences between the crack information and the background of the underwater bridge piers. Compared to the second-place score, the precision, recall, mAP@50, mAP@50–95, and F1 scores are 2 %, 4 %, 2 %, 5 %, and 8 % higher, respectively. Misclassification and failed detection are significantly diminished by UCRT-DETR. Furthermore, its high FPS score demonstrates that it is capable of real-time detection of underwater bridge foundations. The real-time detection of underwater bridge pier cracks by underwater robotics provides a greater engineering utility value, as evidenced by the visualization and operation interface of the upper computer, following practical testing.

In the future, the underwater robot will be equipped with binocular cameras to measure the length and width of the cracks in the underwater bridge piers, multibeam sonar to measure the depth of the cracks, as well as inertial navigation and Doppler range finder to realize the positional localization on the basis of depth localization. Crack detection will be realized while significantly enhancing the assessment of structural

integrity in the underwater environment.

CRediT authorship contribution statement

Zhenming Lv: Writing – review & editing, Writing – original draft, Visualization, Validation, Software, Methodology, Investigation, Formal analysis, Data curation, Conceptualization. **Shaojiang Dong:** Funding acquisition. **Zongyou Xia:** Supervision. **Jingyao He:** Supervision, Investigation. **Jiawei Zhang:** Software.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

The authors do not have permission to share data.

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